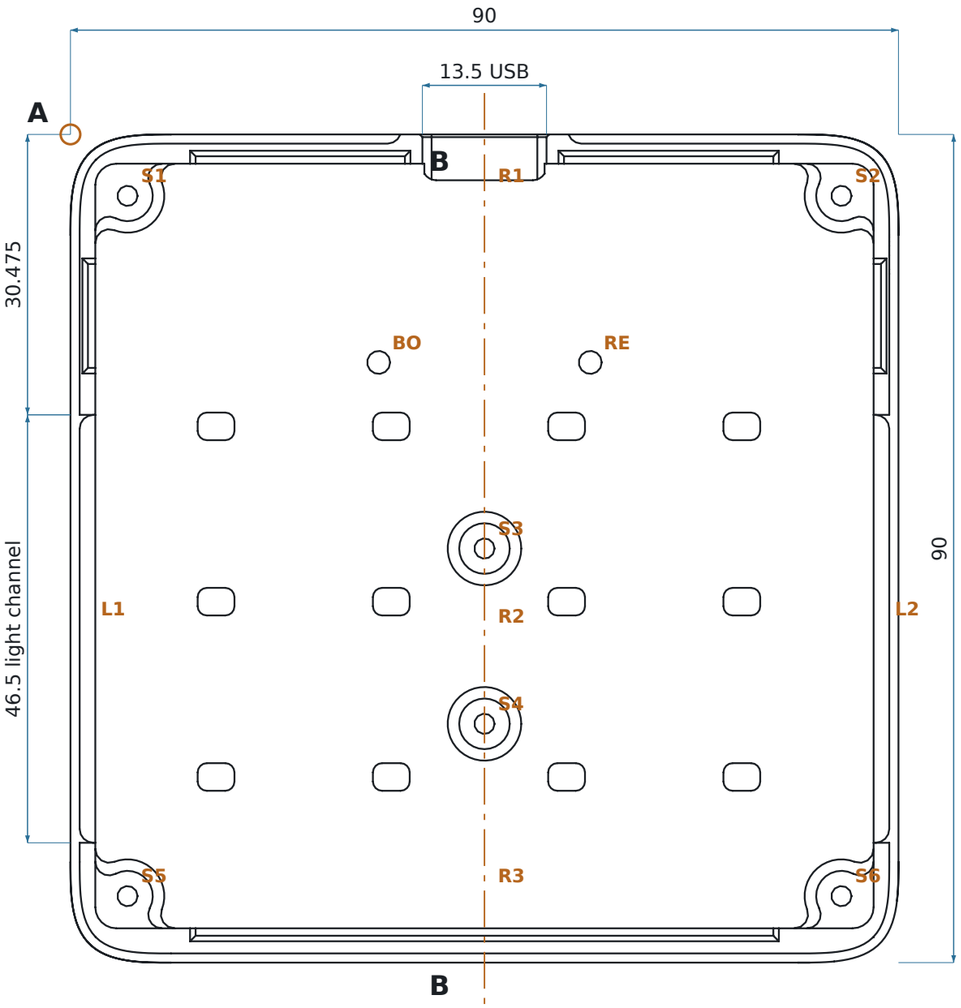
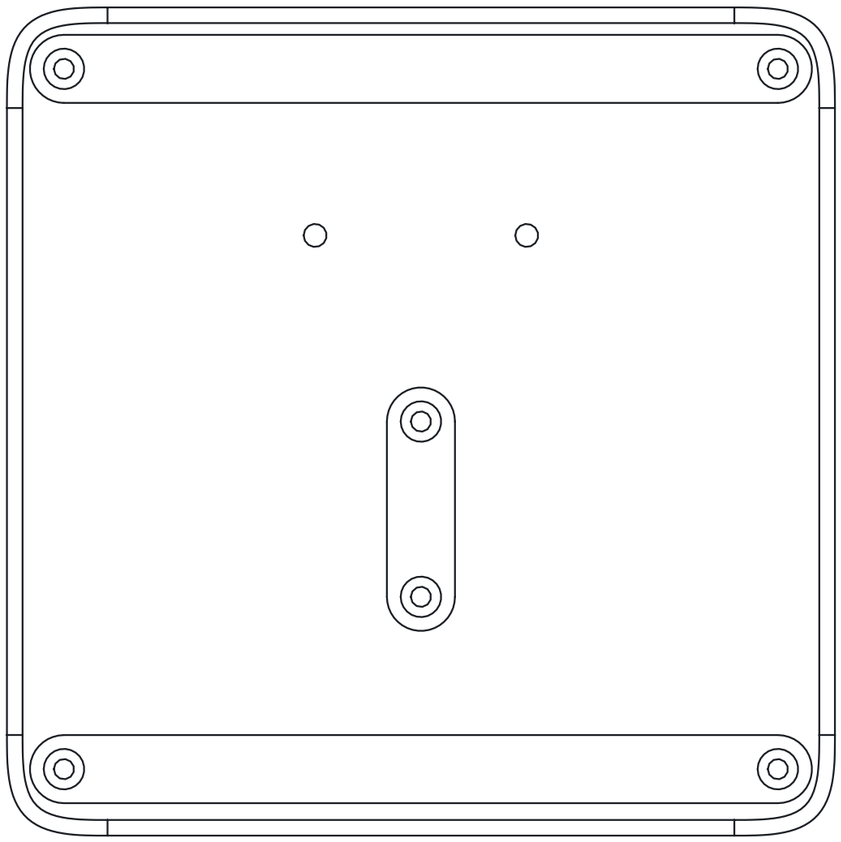
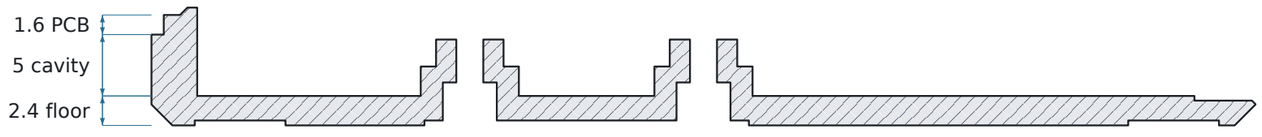
FROM ABOVE

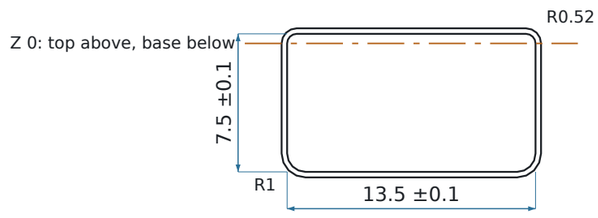


FROM BELOW (mirrored left to right)



SECTION B-B

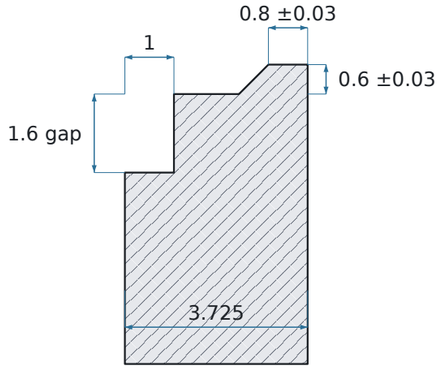




USB OPENING, 3:1

Seen from behind. Rounds along Y: R1 lower (base), R0.52 upper (top). Ball-nose or bull-nose cutter of that radius or smaller, 3D finish; the 3 mm flat end mill cannot cut them. 0.3 x 45 degree lead-in on the outer face.

GAP AND LAP JOINT, 8:1



Scale 1.5:1 plans, 2:1 section, 8:1 detail, on A3. Dimensions in mm, computed from the model.
Datum A: back-left corner of the box round the outline, seen from above. X to the right, Y toward the front.
3-axis, two setups: inside up, then underside up.
Every vertical pocket corner is R1.5 or larger: a 3 mm end mill reaches them all. The USB opening's rounds are not vertical: see its detail.
45 degree chamfers: 1.7 on the underside rim, the flank and ends of the lap ridges, 0.3 on the USB lead-in.
The outer skin's top edge at the gap stays sharp, and so does the rim inside the gap: the top seats on it.
The rim steps in 1 mm over the top 1.6 mm, all round: that is the gap. Cut it with the side of the end mill in setup 1.
Screw holes go straight through: M2 x 8 socket head from below.
Outline corners: one quintic Bezier each, control points 10.94, 6.56 and 2.19 mm from the box corner on both sides. Take the curve from the STEP file.
Laser mark on the underside: see LASER MARK and engraving_underside.dxf. The STEP file is the master for all geometry not dimensioned on this sheet.

PART	macropad case: base, CNC variant
PART NO.	MP-CASE-BASE-CNC REV A
QUANTITY	1 per macropad (order quantity on the purchase order)
MATERIAL	6061-T6 aluminium
FINISH	Bead blast, then clear anodise (natural silver)
UNITS	mm; plans seen from above and from below, as titled
TOLERANCE	ISO 2768-m unless stated; fits as listed below
STATUS	Not made or measured yet: check the first part

Holes and recesses, from datum A

tag	X	Y
⌀2.2 through; counterbore ⌀4.4 to 4 from below; position ±0.05		
S1	6.2	6.675
S2	83.8	6.675
S5	6.2	82.775
S6	83.8	82.775
⌀2.2 through; counterbore ⌀4.4 to 3.5 from below; position ±0.05		
S3	45	45
S4	45	64.05
⌀2.5 through (BOOT and RESET)		
BO	33.5	24.775
RE	56.5	24.775
slot 85 x 7.4, full-radius ends, 0.4 deep from below		
R1	45	6.675
R3	45	82.775
slot 7.4 x 26.45, full-radius ends, 0.4 deep from below		
R2	45	54.525
light channel 46.5 long, 1.7 in, 5.6 deep, R1.5; skin 1.00 ±0.1		
L1	1.85	53.725
L2	88.15	53.725

MATERIAL AND FINISH
6061-T6 aluminium.
Bead blast all over, 120 to 150 grit glass bead, then anodise to MIL-A-8625 Type II, Class 1 (clear, not dyed), 8 to 12 um. Colour: natural silver, as in the renders. Cosmetic faces: the outside and the top face.
About half the coating builds up: each face grows 0.004 to 0.006 mm. The tolerances below hold after anodise.
Break unmarked edges 0.05. The 0.3 chamfers are cosmetic: measure them after blasting.
Rack marks: on the inner cavity walls, below the rim only.
GROUND: mask the counterbore seat of S1 from anodise.
TOLERANCES (after anodise)
General: ISO 2768-m.
S1-S6 positions from datum A: ±0.05, on both parts.
Lap ridges: height 0.6 and crest 0.8, ±0.03 (detail).
USB opening: ±0.1 (detail).
WORKHOLDING
Setup 1: cut the light channels last, after the cavity is roughed: 3 mm 3-flute end mill, 0.2 mm radial steps, full depth in two passes. Each channel leaves a free 1.0 mm skin.
Setup 2 (underside up): sit the base on a plug fixture that fills the cavity, 0.1 mm smaller all round and 4 mm tall, so it backs up both skins. Hold it with an M2 screw through each of the six holes into the fixture, or in soft jaws cut to the outline below the gap. Never in a plain vise.
LASER MARK
Face: the underside, the flat floor face between the foot slots.
Process: fibre laser, after anodise. Fill every closed contour on layer MARK of engraving_underside.dxf (hatch about 0.03 mm, a guess); do not mark layer REF. Colour: the light grey, frosted mark a fibre laser leaves on clear anodise. Mark one sample first.
Position: the DXF is drawn as seen from below, not mirrored, with its origin at datum A. Seen from below with the USB edge away from you, datum A is the top-right corner of the outline box, so a point at (X, Y) in the table above is at (-X, -Y) in the DXF. Register on the REF outline and holes.